

Homework III

Instructions

Show your code and the resulting output. Please ensure that your submission is clearly organized, readable, and easy to follow.

Question 1:

Use Latin Hypercube sample and independent and identically distributed (iid) sample from $U(0,1)$ to approximate the integration of

$$f(x_1, x_2) = \log(1/\sqrt{x_1} + 1/\sqrt{x_2})$$

on $[0, 1]^2$. Note: Let

$$\mu = \int_0^1 \int_0^1 f(x_1, x_2) dx_1 dx_2 = 1.25$$

Compare the estimates and provide brief comments.

Hints: You may follow the steps below:

1. Develop a function that can generate two dimensional Latin Hypercube samples.

```
two_dim_latin_hypercube <- function(n){
  x <- sample(1:n)
  x <- x - runif(n)
  x1 <- x/n
  x2 <- x/n
  cbind(x1, x2) # the value of x1 and x2 are the same here
}

two_dim_latin_hypercube1 <- function(n){
  x1 <- sample(1:n) - runif(n) # this gives big numbers
  x2 <- sample(1:n) - runif(n)
  x1 <- x1/n
  x2 <- x2/n
  cbind(x1, x2)
}

two_dim_latin_hypercube1(10)
```

```
##           x1           x2
## [1,] 0.34155058 0.97979834
## [2,] 0.49511695 0.47348726
## [3,] 0.27050584 0.69763091
## [4,] 0.66124176 0.09318704
## [5,] 0.83038536 0.26259968
```

```
## [6,] 0.51859395 0.58506476
## [7,] 0.98773497 0.12802088
## [8,] 0.17909584 0.77316753
## [9,] 0.08369613 0.80949404
## [10,] 0.72466590 0.30280737
```

2. Develop a function that can generate two dimensional iid uniform samples.

```
# a function that generate two dimensional iid uniform samples
iid_unif_samples <- function(n){
  x1 <- runif(n)
  x2 <- runif(n)
  cbind(x1, x2)
}
iid_unif_samples(10)
```

```
##           x1           x2
## [1,] 0.4283212 0.31387930
## [2,] 0.6682522 0.40373440
## [3,] 0.9344028 0.07953912
## [4,] 0.1385748 0.97536545
## [5,] 0.1788920 0.75085983
## [6,] 0.1549785 0.40684711
## [7,] 0.6729774 0.37844820
## [8,] 0.5769124 0.75970216
## [9,] 0.2795469 0.45356390
## [10,] 0.4687256 0.26313808
```

3. Develop a function that can approximate $\int_0^1 \int_0^1 f(x_1, x_2) dx_1 dx_2$ using a given sample.

```
two.dim.log.fun <- function(x1, x2){
  log(1/sqrt(x1) + 1/sqrt(x2))
}

approx.int.log.fun <- function(x1, x2){
}
```

4. plug in the output of the function in 1 to the function you developed in 3. You can get an approximation of $\int_0^1 \int_0^1 f(x_1, x_2) dx_1 dx_2$ using Latin hypercube samples.
5. plug in the output of function in 2 to the function in 3. You can get an approximation of $\int_0^1 \int_0^1 f(x_1, x_2) dx_1 dx_2$ using iid uniform samples.
6. How far is your approximation from the true value? Which method (lhs and iid) is closer to the true value? If you repeat 4 and 5, will the result change?
7. Replicate 4 and 5 for nrep times. You will have nrep approximations from each method. Let

$$\hat{\mu}_1^{lhs}, \dots, \hat{\mu}_{nrep}^{lhs}$$

and

$$\hat{\mu}_1^{iid}, \dots, \hat{\mu}_{nrep}^{iid}$$

be the resulting approximations. Compute the sample bias:

$$\frac{1}{nrep} \sum_{i=1}^{nrep} \hat{\mu}_i^{iid} - \mu$$

$$\frac{1}{nrep} \sum_{i=1}^{nrep} \hat{\mu}_i^{lhs} - \mu$$

the sample variances:

$$\frac{1}{nrep - 1} \sum_{i=1}^{nrep} \left(\hat{\mu}_i^{iid} - \frac{1}{nrep} \sum_{j=1}^{nrep} \hat{\mu}_j^{iid} \right)^2$$

$$\frac{1}{nrep - 1} \sum_{i=1}^{nrep} \left(\hat{\mu}_i^{lhs} - \frac{1}{nrep} \sum_{j=1}^{nrep} \hat{\mu}_j^{lhs} \right)^2$$

and the mean squared errors:

$$\frac{1}{nrep} \sum_{i=1}^{nrep} (\hat{\mu}_i^{iid} - \mu)^2$$

$$\frac{1}{nrep} \sum_{i=1}^{nrep} (\hat{\mu}_i^{lhs} - \mu)^2$$

8. Try different n . Comment on how is the result change over n .

Your code here

Question 2:

Use simulation to estimate the 90th percentile of the distance between two points (say, (x_1, y_1) and (x_2, y_2)) chosen at random in the interior of a unit square (i.e., $[0, 1]^2$).

Your code here

Question 3:

Use the “Bschoool.csv” dataset to find the answers of the following question within this dataset. Note: Use the pipe operator and functions in tidyverse packages this time.

1. Which professor rank (i.e., associate, assistant, or professor) has the highest average teaching evaluation score?
2. What is the number of female associate professors?
3. Is median teaching evaluation score in fall semester higher than that in spring semester?

Your code here